

Knowledge Alignment Benchmarking: Evaluating Causal Clinical Reasoning in Multimodal LLMs on Brain MRI

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Abstract

Medical multimodal LLMs are usually scored on final-answer accuracy, which cannot show whether a correct answer came from the image, the question text, or a guess. We introduce **Knowledge Alignment Benchmarking (KAB)**, which evaluates brain-MRI reasoning as a sequential five-phase chain and adds vocabulary-and-consistency, feedback-adaptation, and threshold-gating measures. OmniBrainBench’s released groupings cannot support same-patient chains (only 2 verifiable cross-phase patient links among 6,823 questions), so we build a same-patient chain dataset from the LUMIERE glioblastoma cohort. Four image-dependence checks on open-weight models suggest weak image dependence: text-only accuracy is within about ± 8 points of image-present accuracy (a margin set post hoc), stated evidence mostly repeats shown text, and answers that change under a substituted image match the substituted patient’s label only modestly above item-specific chance (34% vs. 22%). Answerability from a single slice and the answer keys are not yet clinician-validated.

1 Introduction

Multimodal large language models (MLLMs) are increasingly applied to medical imaging, yet the dominant evaluation treats medical visual question answering like any multiple-choice benchmark and scores only the selected option. Final-answer accuracy cannot separate a model that read the image and reasoned validly from one that pattern-matched cues in the question or options, or guessed. Such shortcuts are well documented outside medicine: hypotheses alone suffice to beat chance on natural language inference (Gururangan et al., 2018), models adopt syntactic heuristics that fail under adversarial evaluation (McCoy et al., 2019), and stated chain-of-thought often does not reflect what produced the answer (Lanham et al., 2023; Arcuschin

et al., 2025; Somov et al., 2026). In medical multimodal models, early reasoning failures cascade into later errors (Jung et al., 2026), and models rely on textual or visual shortcuts rather than the pathology a question asks about (Nguyen et al., 2025). Clinically, an accurate answer that does not depend on the image cannot be trusted on the next case that differs slightly.

We introduce **Knowledge Alignment Benchmarking (KAB)**, which evaluates a brain MRI case as a sequential chain of five clinical phases in which later judgments build on earlier ones (Croskerry, 2009): Anatomical and Imaging Assessment (AIA), Lesion Identification and Localization (LIL), Diagnostic Synthesis and Causal Reasoning (DSCR), Prognostic Judgment and Risk Forecasting (PJRF), and Therapeutic Cycle Management (TCM). Beyond accuracy, KAB scores whether a model’s reasoning uses valid clinical vocabulary and is self-consistent (**vocabulary and consistency**), whether it can use a targeted clinical correction (**feedback-adaptation**), and whether its answers are jointly correct across phases (**threshold gating**, a scoring rule rather than a demonstrated dependency). Figure 1 summarizes the framework.

Building KAB on OmniBrainBench (Peng et al., 2026), we found that the released benchmark cannot support the same-patient sequential evaluation KAB requires: its `source_file` grouping, which we had interpreted as patient identity, does not track it. Of 6,823 questions, only 2 patients (4 questions) form a verifiable cross-phase pair, a lower bound since verification relies on shared images or filename-derived identifiers (Section 4). This is an incompatibility between the released benchmark and our evaluation; we are not aware of an explicit claim by OmniBrainBench that its questions form patient-linked chains. We therefore build a same-patient chain dataset from LUMIERE (Suter et al., 2022), a 91-patient glioblastoma cohort with lon-

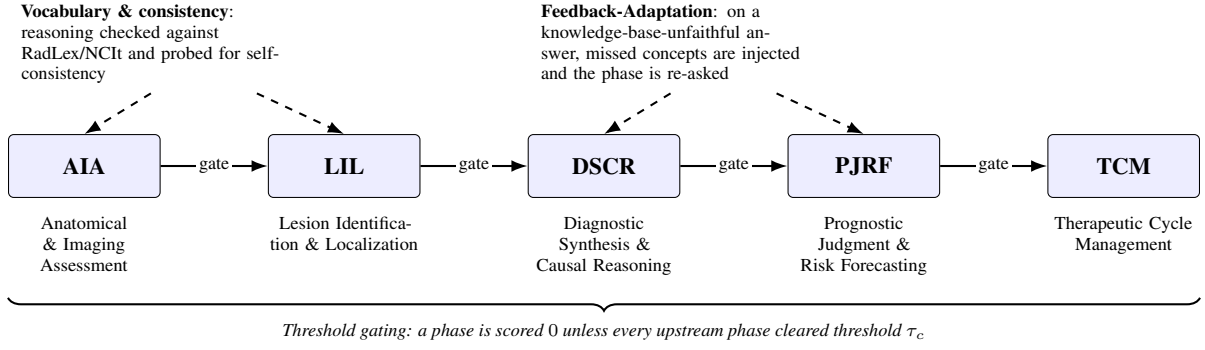


Figure 1: The KAB framework: a brain-MRI case is evaluated as a five-phase sequential chain (bottom), with three metrics attached to it (top and below): vocabulary and consistency (ontology term validity + answer-reasoning self-consistency), feedback-adaptation (a targeted re-query on failure), and threshold gating (a phase counts only if its upstream phases cleared a score threshold).

itudinal imaging, expert RANO ratings, survival, and treatment records, using LLM drafting and an automatic leakage auditor (Section 5).

This paper makes three contributions:

- The KAB framework: ontology-based vocabulary scoring with an answer-reasoning consistency probe, feedback-adaptation, and threshold gating. Adaptation and gating are exploratory: adaptation triggered on only 1–4 of 52 patients per model, and the gating-causality experiment (Section 7) has not reported results.
- A documented incompatibility of an existing brain-MRI benchmark with same-patient sequential evaluation, and a reusable method for building a same-patient chain dataset from a longitudinal clinical cohort.
- Four image-dependence checks on four open-weight models (Section 7), whose results point to weak image dependence. Answerability and answer keys are not clinician-validated (Section 8), and proprietary models are not yet run.

2 Related Work

Shortcuts and reasoning faithfulness. A model can reach a correct label without doing the intended task: natural language inference hypotheses alone carry enough annotation artifacts to outperform chance (Gururangan et al., 2018), and models trained on such data adopt syntactic heuristics that fail on HANS (McCoy et al., 2019). For chain-of-thought, Lanham et al. (2023) test whether stated reasoning causally affects the answer and find that it often does not; Arcuschin et al. (2025) report unfaithfulness up to 13% on natural prompts in production models; Somov et al. (2026) find that

models fail to update their decision after intervention on intermediate structures in up to 60% of cases. KAB’s answer-recoverability probe is closest to this line but tests only textual consistency: we withhold everything except the reasoning text and ask whether it reproduces the answer.

Cascading failure in medical reasoning. Jung et al. (2026) propose a training method that penalizes earlier invalid reasoning steps more heavily, targeting the cascading failure that motivates KAB’s gating; Nguyen et al. (2025) introduce perturbation tests for textual and visual shortcuts in medical VQA. Neither asks whether the multi-step cases represent one patient’s trajectory, which the first presupposes. The five-phase structure follows dual-process accounts of diagnosis in which sequential steps characterize diagnostic error (Croskerry, 2009), though we treat the dependency between phases as an assumption to be tested (Section 7).

Brain MRI benchmarks and same-patient continuity. OmniBrainBench (Peng et al., 2026) is, to our knowledge, the largest closed-ended brain-MRI MLLM benchmark organized around a multi-phase workflow, and is where we started. NeuroQA (Abbasi et al., 2026) (56,953 QA pairs, 12,977 subjects) has a longitudinal category that compares two timepoints, but its authors state it does not support claims about treatment planning; UCSF-PDGM-VQA (Ghosh et al., 2026) (2,387 pairs, 473 glioma studies) has one timepoint per patient and no prognosis or treatment layer. LUMIERE and MU-Glioma-Post (Washington University School of Medicine et al., 2025) provide real longitudinal data with treatment and outcomes but no question

layer. We are not aware of a published dataset combining same-patient continuity from imaging to treatment with a multiple-choice layer; Section 5 builds one on LUMIERE.

3 The KAB Framework

KAB evaluates a model per case, a case being one patient’s progression through the five phases. For each question the model receives the image(s), question, options, and a summary of its own earlier answers in the case (the *chain context*), and returns an answer, a description of what it observed in the image (*visual_grounding*), and reasoning. KAB computes three families of measures on top of this interaction.

3.1 Vocabulary and Consistency Metrics

Answer recoverability: a second call sees only the model’s reasoning text and asks which answer it supports; a match means the reasoning is textually sufficient for the answer. This establishes self-consistency, not that the reasoning caused the answer. *Ontology term coverage*: the fraction of the response’s knowledge-base terms that belong to the phase-appropriate base (RadLex for AIA and LIL, the NCI Thesaurus for DSCR, PJRF, and TCM); it measures term validity, not correctness. We also compute answer-term overlap and upstream-term repetition (Appendix C). We call these vocabulary and consistency measures because none tests causal faithfulness.

3.2 Feedback-Based Adaptation

When ontology coverage falls below $\tau_g = 0.3$ and missed concepts exist (phase-KB terms in the correct answer but absent from the response), we re-ask with the model’s prior answer and a correction block listing the missed concepts, and report the fraction corrected. This measures performance under a targeted, answer-derived hint. Because the hint supplies new information, it does not separate a knowledge gap from an activation failure; that would need generic-retry and irrelevant-hint controls, which we have not run.

3.3 Threshold Gating

A phase’s score is set to 0 (*gated_out*) unless every phase it is assumed to depend on scored at least $\tau_c = 0.5$: LIL requires AIA; DSCR requires AIA and LIL; PJRF the first three; TCM all four. The rule is binary, so it is hard gating. It is an imposed scoring convention, not a demonstrated

dependency: a model can misidentify a sequence yet localize a lesion, or answer every phase through independent shortcuts, and our DSCR answer keys do not depend on the LIL baseline (Section 5). We test the dependency by manipulating upstream context (Section 7). In the implementation a blocked phase is not queried, but a queried phase’s inputs do not depend on gating, so a gated run equals a non-gated run with blocked scores set to zero; all results here use non-gated runs. A high chain completion rate (every phase ran and cleared its gate) means above-threshold scores at every phase, not necessarily coherent reasoning.

4 Compatibility of OmniBrainBench with Sequential Evaluation

KAB presupposes that a case’s phases are one patient’s trajectory. We tested whether OmniBrainBench supports this by checking whether any image is shared across questions of different phases within a grouped case; shared images are the continuity evidence we can verify directly, though not the only conceivable evidence. Of its 6,823 closed-ended questions (15 source sub-corpora), only 2 verifiable cross-phase patient links exist, both in VQA_RAD (one AIA and one LIL question each, 4 questions), and none span three or more phases. A second method, extracting per-patient identifiers from filenames corpus by corpus, gave the same result. The release provides no complete patient identifier, so these are lower bounds, not a census. The cause is structural: each sub-corpus was built for one task (BraTS for tumor characterization, RSNA for hemorrhage, ISLES for stroke), so a patient is essentially asked one phase-type of question. We had interpreted *source_file*, an origin annotation-batch field, as patient identity; that was our KAB-side choice, and under it the grouping stitches together different patients’ questions. We frame this as an incompatibility between the released benchmark and our evaluation, not a defect in a promise its authors made (Appendix D).

5 A Same-Patient Chain Dataset from LUMIERE

LUMIERE (Suter et al., 2022) has 91 glioblastoma patients, 638 study dates, automated multi-sequence segmentation, and expert RANO ratings with rationale. Figure 3 (Appendix B) summarizes the pipeline.

Facts. Per-patient facts are extracted with provenance from LUMIERE’s shipped data: segmentation masks already registered to MNI152 space reduce lesion location to a voxel lookup, and volumes ship precomputed. DSCR uses the released expert RANO ratings directly; we did not re-derive or audit them. RANO assessment (Wen et al., 2010) draws on more than an image pair (reference scan, clinical status, corticosteroid dose) and the model receives none of the latter, so a DSCR answer cannot be assumed derivable from the model-visible inputs. PJRF and TCM facts come from the survival and Stupp-protocol records.

Cohort and drafting. Of 91 patients, 55 meet the eligibility criterion (at least 3 RANO response-rated follow-ups and at least one timepoint with complete multi-sequence imaging); one has no timepoint with both and is dropped, leaving 54 patients and 269 drafted items. Claude 4.5 Sonnet drafts one multiple-choice item per phase with distractors from the facts. AIA, LIL, and DSCR are nearly programmatically derivable; PJRF and TCM need distractors informed by the glioblastoma prognosis and treatment literature (for example pseudoprogression), which the drafting prompt encodes.

Item construction. For each patient we sample one random eligible timepoint that has both an expert RANO rating and complete imaging (fixed seed); always taking the last rated timepoint would make nearly every DSCR answer progressive disease. The resulting DSCR labels over the 54-patient cohort are 37 progressive disease, 9 stable disease, 4 partial response, and 4 complete response. The LIL volume change is measured from each patient’s latest post-operative scan before the follow-up, so that it reflects treatment response rather than resection; two patients without a matching post-operative segmentation are excluded, leaving 52 patients. DSCR, PJRF, and TCM answer keys do not depend on the LIL baseline. Drafted items pass through an automatic stem-leakage auditor that flags any stem or option restating the phase’s own answer or an upstream phase’s answer (a percent volume change, region name, change-direction word, RANO label, or survival duration); violations are fed back into the drafting prompt for up to eight retries, and the DSCR stem is built from a fixed template of structured fields with no answer information available to leak. The resulting item set (v3) has zero violations in every phase, which

shows compliance with that auditor, whose rules the drafting loop was optimized against, not the absence of leakage. The development history (two sampling bugs, the leakier initial set v2, and the rebaselining) is in Appendix B.

Image rendering. Each timepoint is one 2D axial slice: the slice with the largest tumor-mask area from the atlas-registered contrast-enhanced T1, rotated to radiological orientation and clipped to the 1st–99th percentile before scaling to 8-bit grayscale. There is no multi-slice or 3D rendering, and for the leak-controlled set the prompt states the radiological display convention. This bears on answerability: LIL asks about whole-lesion volume change computed from a 3D segmentation, which two maximum-area slices do not determine, so low LIL accuracy cannot be attributed to failure to use available evidence. No clinician has yet judged whether each question is answerable from exactly the images and text shown (Section 8).

6 Experimental Setup

Models. Four open-weight models served through Ollama at 4-bit (Q4_K_M): MedGemma-4B (the only medical fine-tune), Gemma-3-12B, Gemma-3-27B, and Llama-4-Scout (109B mixture-of-experts, 17B active); tags medgemma:4b, gemma3:12b, gemma3:27b, llama4:scout, each on one RTX PRO 6000. Proprietary models (GPT-5, Claude 4.5 Sonnet, Gemini 2.5 Pro) are planned but not yet run. All models use one OpenAI-compatible interface with greedy decoding (temperature 0) and an 800-token budget, and the response schema asks for answer before grounding and reasoning so a truncated response still has an answer.

Failed responses. A call is retried up to 3 times with exponential backoff. If the JSON fails to parse (truncation, or unescaped newlines, recovered by a lenient parse), we salvage the answer letter by regex; if that fails the item is scored incorrect, not excluded, so a model prone to malformed output does not get an easier denominator. Salvaged answers are scored on the salvaged letter. (The evaluator’s per-run report instead drops parse-error items from a phase’s denominator, which lowers MedGemma-4B’s image and text-only accuracy from 34.2% and 36.2% to 33.1% and 35.8% (3 and 1 items); we report the per-question basis.)

Knowledge bases. RadLex (RSNA OWL, downloaded 2026-08-05) and the NCI Thesaurus (OWL v26.07d) are used as-is; the aligner raises an error if either is missing.

7 Testing Whether Correct Answers Are Image-Grounded

The vocabulary and consistency metrics test whether stated reasoning is internally consistent, not whether it derives from the image; recoverability in particular passes any response whose reasoning restates its chosen option. We therefore run four checks on the leak-controlled item set, identically across models.

Stem-leakage audit. The auditor of Section 5 doubles as a diagnostic: an item whose stem or upstream context already states the answer tests nothing about image use. Running it on both item sets is a controlled manipulation for the other checks.

Text-only ablation. We strip all images and compare accuracy with the image-present run on the same items. Both arms are non-gated so every phase is scored. Chain context is regenerated from each arm’s own earlier answers, so later-phase differences combine a same-phase effect with propagated upstream effects. If the patient-clustered paired difference lies within an equivalence margin, the image is not necessary for the answer on this item set; a nonsignificant difference alone would not show this.

Evidence fact-check. For each checkable claim in a response’s `visual_grounding` and reasoning (hemisphere or region statements, and measurements with units) we record two properties: *availability* (already in the stem, options, or upstream context shown) and *record status* (supported, contradicted, unmatched, or unverifiable against the structured facts). Responses with no checkable claim are reported separately. The audit is deterministic pattern matching and does not interpret paraphrase (Appendix E).

Counterfactual-image substitution. We replace each patient’s image with a different real patient’s (the *donor*’s), keeping every text field, including timepoint labels, unchanged. A deterministic seeded pairing gives the donor an opposite-sign lesion-volume change. LIL and DSCR are the primary phases; AIA’s answer is a fixed string and serves as a reference; PJRF and TCM mix same-

phase and propagated upstream effects. The options remain the original patient’s, so the donor’s fully correct answer is generally absent and the donor’s clinical facts in the stem are not substituted; truth-tracking is testable only at label level (volume-change direction for LIL, RANO category for DSCR). The outcomes are the flip rate and, among flips, agreement with the donor’s label judged against an *item-specific* null: the probability that a uniformly chosen other option carries the donor’s label.

Repeated identical-input control. For each model we re-run the own-image condition with identical inputs and greedy decoding (run tag rep2) to measure baseline answer variability. These runs are queued and not yet reported, so we do not attribute any answer change to decoding variability.

Gating-causality experiment. To test rather than impose the dependency between phases, we evaluate LIL, DSCR, PJRF, and TCM under three manufactured upstream-context conditions (correct upstream answers, a fixed incorrect option per upstream phase, no upstream context) with image, question, and options held fixed. If downstream accuracy does not differ across conditions, the assumed dependency is not supported. These runs are queued and not yet reported.

Analysis plan. Before running the v3 and counterfactual jobs we recorded a joint decision rule in our dated project log (paper/update.md, 2026-09-22): text-only accuracy within the confidence interval of image-present accuracy, evidence mostly restated from shown text, and a low, non-truth-tracking flip rate together indicate non-grounding; the opposite pattern indicates grounding. The log was not externally timestamped, so this is a pre-specified analysis plan, not a preregistration. It used confidence-interval overlap; the $\pm 8\text{pp}$ margin below was adopted afterwards, after we had seen the intervals, and describes what the data can exclude rather than a prespecified test.

8 Results

All results use the leak-controlled (v3) item set and the four open-weight models. The initial item set (v2) failed the stem-leakage auditor at rates too high to treat its accuracy as informative about image use (Appendix B); v3 clears every phase with zero violations, which shows compliance with that auditor, not that no leakage remains.

Model	Image	Text-only	Δ [95% CI]
Llama-4-Scout	48.8	51.9	-3.1 [-7.3, +1.2]
Gemma-3-27B	53.5	50.4	+3.1 [-0.8, +7.3]
Gemma-3-12B	48.1	50.8	-2.7 [-7.3, +2.3]
MedGemma-4B	34.2	36.2	-1.9 [-5.8, +1.9]

Table 1: Non-gated accuracy (%) with images versus text-only, and the paired difference $\Delta = \text{image} - \text{text-only}$ in percentage points with its patient-clustered 95% bootstrap interval (52 patients, 10,000 resamples). Accuracy is per question (Section 6).

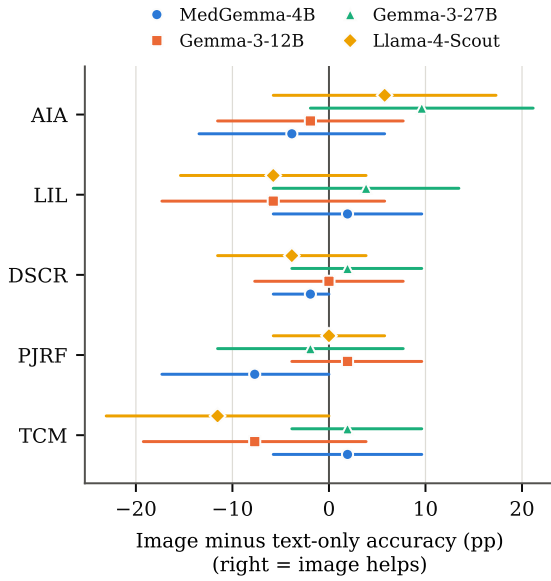


Figure 2: Paired accuracy difference (image minus text-only, percentage points) by phase and model, with patient-clustered 95% bootstrap intervals; right of zero means the image helped. Not adjusted for multiplicity across the 20 cells.

Text-only ablation. Table 1 gives accuracy with and without images and Figure 2 the paired per-phase differences; patients, not questions, are resampled because each contributes one question per phase. Three of four models score at least as high without the image (image minus text-only: Llama-4-Scout -3.1pp, Gemma-3-12B -2.7, MedGemma-4B -1.9). Gemma-3-27B is the exception (+3.1), concentrated in AIA (+9.6pp, [-1.9, +21.2]; 8 items right only with the image, 3 only without). Every overall interval includes zero; all fall within ± 8 pp and none within ± 5 pp. We chose the ± 8 pp margin after seeing them, so it describes what the data exclude and is not a prespecified test, and 52 patients give limited power against a moderate effect. Across the 20 model-phase cells no interval lies entirely

Model	AIA	LIL	DSCR
MedGemma-4B	17 [8, 29]	12 [4, 21]	6 [0, 13]
Gemma-3-12B	10 [2, 17]	37 [23, 50]	27 [15, 38]
Gemma-3-27B	17 [8, 29]	42 [29, 56]	25 [13, 37]
Llama-4-Scout	10 [2, 17]	23 [12, 35]	12 [4, 21]

Table 2: Flip rate (%) between own-image and donor-image runs with patient-clustered 95% bootstrap intervals ($n = 52$ per cell). AIA’s answer does not depend on the image.

on the image-advantage side. A model gives the same answer with and without the image on 77–89% of items and the same correct-or-incorrect outcome on 87–91%; among items whose outcome differs, right-only-with-image and right-only-without-image are comparable (55 versus 67), which is consistent with decoding variability but not evidence for it, pending the repeated-run control.

Evidence fact-check. 26–55% of answers contain no checkable claim (MedGemma-4B 35%, Gemma-3-12B 31%, Gemma-3-27B 55%, Llama-4-Scout 26%); they are excluded from the claim-level denominators below. Among the remaining claims (251–668 per model), 75–87% were available in shown text and match the record; 10–18% were available in shown text yet contradict or fail to match the record (for example a wrong location repeated from an earlier turn, which a single “copied” label would hide); only 1–2% were absent from shown text and match the record; and 2–6% were absent and contradict or fail to match. The small share of novel, record-matching claims means the audit finds little evidence that stated evidence contains facts obtainable only from the image. It cannot show the converse: a model can inspect an image and choose wording that also appears in an option, and a novel claim that matches the record need not have been read from the image. The audit is descriptive, misses paraphrase, and is not validated against expert annotation.

Counterfactual-image substitution. Table 2 gives flip rates. AIA’s flip rate (10–17%) is not a clean decoding-variability baseline, since an image change may still perturb a response; the direct control is not yet reported. LIL (12–42%) and DSCR (6–27%) flip more often than AIA for Gemma-3-12B and Gemma-3-27B on both phases and for Llama-4-Scout on LIL, and no more often for MedGemma-4B; several intervals overlap AIA’s. PJRF (4–23%) and TCM (10–37%) also flip,

but their upstream context is regenerated under the swapped image.

Among the 95 flipped LIL and DSCR items, 32 (33.7%) carry the donor's label. If each flipped answer were a uniformly chosen other option, the expected agreement is 20.7 of 95 (21.8%); the observed rate exceeds this item-specific null by 11.9pp (patient-clustered 95% interval $[+3.5, +20.3]$; donor-clustered $[+2.9, +20.7]$). By phase, LIL is 18 of 59 (12.0 expected) and DSCR 14 of 36 (8.7 expected). This is modest evidence of some image influence, not of grounding: 30.5% of flips keep the same direction or RANO label and cannot track the swap, and in LIL 67.8% of flips land on the original patient's own direction (47.2% in DSCR, where the two patients' labels sometimes coincide), consistent with the text dominating. The check is limited by option availability (the donor's correct LIL option was among those shown for 0 of 52 patients, and its direction label for 37 of 52; for DSCR, 48 and 52 of 52), by donor facts in the stem that were not substituted, by shared patients and reused donors (addressed by clustered intervals, not a donor permutation), and by 2 of 52 pairs in which one patient's LIL truth is a near-zero, "stable" change.

Other measures. Answer recoverability is 96–100% for every model and phase, so reasoning almost always predicts the answer; this is internal consistency, not image derivation. Ontology coverage is bimodal (15–48% on AIA and LIL, 94–98% on DSCR, PJRF, and TCM), reflecting each phase's answer vocabulary rather than reasoning quality. Adaptation triggered on 1–4 of 52 patients per model, only in LIL, too sparse to conclude anything, and gated runs are not reported because all v3 runs used gating disabled. Two option-only observations qualify accuracy (Appendix E): always answering *progressive disease* scores 67% on DSCR (typically also the longest option), above every model (65, 58, 56, and 14%), so no model shows RANO skill beyond class imbalance; and LIL accuracy is at or near the 25% chance level for three of four models (23, 27, 25%; MedGemma-4B 40%), with text-only LIL accuracy statistically the same (23–38%). The LIL result cannot be read as an inability to extract volume-change direction, because one slice per timepoint does not determine total-volume change.

Reading against the analysis plan. The first two conditions are met: no model shows an image ad-

vantage whose paired interval excludes zero, all four overall intervals sit within a ± 8 pp margin (adopted after inspection; not within ± 5 pp), and novel, record-matching claims are 1–2% of checkable claims. The third, a low and non-truth-tracking flip rate, is not met: LIL and DSCR flip rates are not low, and flipped answers agree with the donor's label modestly more often (33.7%) than the item-specific null (21.8%). On this item set and these models, accuracy therefore does not depend measurably on the image, and answers are somewhat sensitive to the image without tracking it reliably. We do not conclude that the images are ignored or that the models cannot use them: answerability from the supplied slice is unvalidated, the flip result awaits the repeated-run control, and answer keys have not been checked by a clinician.

Limitations

The most consequential limitation is that no clinician has validated the item set as answerable. Each timepoint is one 2D slice, yet LIL asks about whole-lesion volume change and DSCR about a RANO category whose definition draws on reference-scan choice, clinical status, and corticosteroid dose that the model is not shown. Low accuracy on these phases may therefore reflect insufficient input rather than failure to use available evidence, and the finding that answers do not depend on the image has the same ambiguity in reverse: an item that cannot be answered from the image will not show image dependence. PJRF and TCM keys are defined by realized outcome (the survival bucket consistent with recorded overall survival) and documented treatment, not by a forecast or recommendation validated as justified at a stated prediction time from the information then available; a plausible alternative forecast is scored wrong, so these phases measure agreement with the record. Distractors were LLM-drafted and their quality and label validity are unreviewed. An external clinical reviewer would address these questions and supply the expert annotation needed to validate the evidence audit; that review has not been done.

The results cover the four open-weight models only. The item set is a correction of a leakier earlier draft, and zero auditor violations show compliance with an auditor the drafting loop was optimized against, not absence of leakage. The equivalence margin for the text-only comparison was set after seeing the intervals, and the counterfactual check

did not meet the analysis plan’s “low, non-truth-tracking flip rate” condition. Intervals treat patients as exchangeable, are not adjusted for multiplicity across the 20 model-phase cells, and share the same 52 patients and donor pairing across models. Ontology term coverage is a string-matching metric against fixed vocabularies that measures term validity, not correctness, and it undercounts correct concepts phrased outside the ontologies; answer recoverability detects only inconsistency between reasoning and answer. The evidence audit is deterministic pattern matching without paraphrase handling and is unvalidated. In OmniBrainBench, PJRF and TCM are 64 and 51 questions, so results there need confidence intervals and must not be pooled; LUMIERE’s per-phase intervals are wide at $n = 52$, and we report them rather than one pooled number. LUMIERE’s tumor segmentations are automated (DeepBraTumIA, HD-GLIO-AUTO) and not manually verified, whereas RANO ratings are expert-assigned, and LUMIERE is a single-center cohort. Finally, no model here needs multi-GPU tensor parallelism, so a larger open-weight model would reopen that question.

Ethics Statement

This work evaluates the diagnostic reasoning of general-purpose and medical-domain language models on real, de-identified patient imaging data; it does not deploy, recommend, or endorse any model for clinical use, and no results in this paper should be interpreted as a clinical validation of any evaluated system. LUMIERE is used under its published non-commercial research license, and all patient data is accessed in its already de-identified, publicly released form; no new identifiable patient information is collected.

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A A Worked Patient Example

To make the abstract description of items, chain context, and evidence auditing in Sections 3-8 concrete, this appendix walks through one complete case end to end: Patient-002’s leak-controlled (v3) five-phase chain, as answered by Gemma-3-27B (the best-performing open-weight model), together with the evidence fact-check’s actual (audit-tool-generated, not hand-picked) classification of the model’s stated reasoning at each phase.

The patient. Patient-002 is a 71-year-old female with glioblastoma (MGMT unmethylated, IDH status unavailable) who underwent complete resection of the enhancing tumor followed by Stupp-protocol chemoradiation. The post-operative re-baselined LIL comparison is week-040-2 (baseline) versus week-047 (follow-up): total lesion volume increases from 27,937mm³ to 57,570mm³ (+106%), with the enhancing core and necrotic components moving from the Superior Parietal Lobule at baseline to the Left Cerebral White Matter at follow-up. The week-047 RANO rating is progressive disease. Recorded overall survival from surgery is 48 weeks.

AIA. *Question:* which imaging modalities were available. *Correct/model answer:* T1, contrast-enhanced T1, T2, and FLAIR (correct). The model’s stated visual_grounding for this item, unprompted, also volunteers a lesion location: “a mass lesion in the right temporal lobe.” No part of the AIA question or options mentions location, so this claim is neither asked for nor copied from anything shown to the model. Checked against the same structured facts used to draft the item, it is

wrong on two counts: the true baseline location is the Superior Parietal Lobule/Left Cerebral White Matter (not temporal), and the true laterality is left, not right. Running the evidence fact-check’s hemisphere check on this claim in isolation (Section 7) labels it *laterality_flip*: a real, audit-caught contradiction, reported under that label rather than as ordinary fabrication per the paper’s own convention (Section 7), since a display-orientation ambiguity cannot be ruled out for this claim in the same way it can for later ones below.

LIL. *Question:* how did the lesion’s location and volume change between baseline and follow-up. *Correct/model answer:* shifted predominantly to the Left Cerebral White Matter, +106% (correct). The model’s reasoning restates the same fabricated baseline location from AIA (“the initial lesion is located in the right temporal lobe”), now unchanged across a full phase boundary, and concludes with the correct answer’s own location and percentage. Run through the fact-check, every checkable claim in this response, the repeated “right temporal lobe” and the “106%” figure alike, is labeled *copied* under the earlier single-label scheme: not because the audit missed the AIA-level contradiction, but because by LIL, both claims are present in the model’s own prior-turn output or the shown answer options, which is why that label cannot distinguish a genuinely re-verified observation from one simply carried forward. Under the two-dimension scoring of Section 7 the same response contains 3 claims that are available in shown text and match the record and 2 (the repeated “right temporal lobe”) that are available in shown text but contradict it, so the carried-forward error remains visible.

DSCR. *Question:* RANO response category at week-047. *Correct/model answer:* progressive disease (correct). The model’s evidence again repeats the “right temporal lobe” framing and the radiology-report measurement it was shown, and the fact-check again labels every claim *copied*. The final answer is correct, and consistent with the (still-wrong) location claim the model has now repeated across three consecutive phases without correction.

PJRF. *Question:* expected overall survival from surgery. *Correct answer:* approximately 12 months (48-52 weeks) — age over 70 as the dominant adverse factor despite complete resection. *Model answer:* approximately 9 months (36-40 weeks) — the option attributing a *particularly* poor prognos-

sis to the same unmethylated-MGMT/age-over-70 combination. This is the one incorrect answer in the chain, but it illustrates a limit of the answer key rather than only a model error: both options describe the same adverse risk factors in the same direction, the key is the bucket consistent with the *realized* 48-week survival, and a nine-month forecast is a plausible prognosis for a 71-year-old with unmethylated MGMT that the key marks wrong. The example also shows why the information cutoff matters: the follow-up imaging is at week 47 and survival is measured from surgery, so the outcome falls one week after the last image.

TCM. *Question:* appropriate management at this time. *Correct/model answer:* initiate second-line therapy given evidence of true progression (correct).

Reading this case. Taken as a whole, this single chain illustrates the gap Section 8 reports in aggregate: 4 of 5 final answers are correct, but the evidentiary trail behind the LIL and DSCR answers rests on a spatial claim introduced at AIA that the fact-check itself flags as contradicting the ground truth, not on independently re-derived observation at each phase. We did not select this patient for this property after the fact; it is Patient-002, the first case in the evaluation order, and its fact-check labels above are exactly what `tools/audit_reasoning_facts.py` outputs when run on it, not a paraphrase.

B Item-Set Development History

This appendix records the construction problems found and fixed while building the LUMIERE item set, in the order they were found. Section 5 describes only the final procedure; Figure 3 summarizes it.

Timepoint sampling. We initially selected each patient’s most recent RANO-rated timepoint, but found this made nearly every DSCR answer *progressive disease*, since glioblastoma patients overwhelmingly trend toward progression by their final scan, a sampling choice that would have made the DSCR phase trivially guessable regardless of a model’s actual reasoning. We instead sample a random eligible response-rated timepoint per patient (fixed seed), which yields a substantially more balanced label distribution across the 54-patient cohort: 37 progressive disease, 9 stable disease, 4

partial response, and 4 complete response. We additionally found and fixed a related bug in which a randomly sampled RANO-rated timepoint could lack matching complete imaging, silently falling back to imaging from a *different* date than the one the RANO rating and DSCR facts describe; we now constrain sampling to timepoints with both a rating and complete imaging, and verified zero such mismatches across the full extracted cohort.

Stem-leakage audit and a leak-controlled item set. Manual inspection of the drafted items surfaced a systematic construction flaw: the drafter wrote each phase’s question stem as a self-contained clinical vignette, which led it to restate findings established in earlier phases of the same chain. We built an automatic auditor that flags, per phase, any stem or option text containing content it should not have access to: a numeric percent volume change, a raw volume or region name, a change-direction word, a RANO label, or a stated survival duration, checked both against the phase’s own correct answer (*self-leakage*) and against the correct answers of upstream phases in the same chain (*upstream leakage*). Applying this auditor to the initial item set (v2) found leakage in a majority of DSCR, PJRF, and TCM items and a smaller fraction of LIL items; a lexical baseline that guesses the option with the highest word overlap with the stem performs at or below chance on the same items, confirming the leak is semantic (e.g., a stated percent volume increase implying *progressive disease*) rather than a copy-paste artifact a naive overlap check would already catch.

We traced the cause to the drafting prompt itself, which supplied each phase’s drafting call with the correct answers of upstream phases (necessary for constructing a coherent chain) without instructing the model not to restate them, and, for DSCR specifically, explicitly asked the model to let the reader *infer* the response label from volume-change facts it had just been shown. We built a revised, leak-controlled item set (v3) that keeps the same upstream-truth context available to the drafter but forbids restating it in the visible stem or options, re-drafting automatically (up to eight attempts) whenever the auditor flags a violation and feeding the specific violated rule back into the retry prompt. The DSCR stem, which the auditor showed was hardest to keep leak-free under free-form drafting, is instead built from a fixed template populated only from structured extent-of-resection, re-

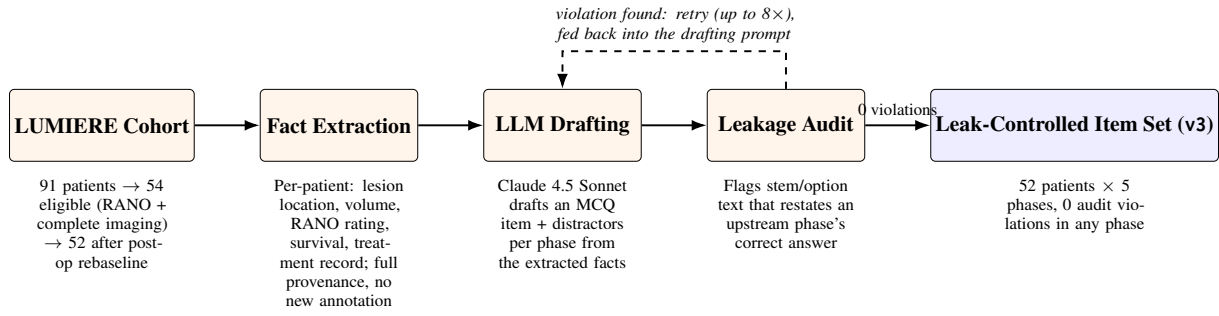


Figure 3: Construction pipeline for the leak-controlled (v3) LUMIERE item set. Structured per-patient facts are extracted from LUMIERE, an LLM drafts multiple-choice items from those facts, and an automatic auditor flags any item that restates an upstream phase’s correct answer, feeding violations back into the drafting prompt for up to eight retries.

resection, and radiotherapy-timing fields, with no correct-answer information available to leak. The leak-controlled item set clears the auditor with zero violations across every phase. This establishes compliance with that auditor, whose rules the drafting loop was itself optimized against, not the absence of leakage.

Stem-leakage rates on v2. The initial item set (v2) failed the auditor at rates too high to treat its accuracy numbers as informative about image use: 0% of AIA items leaked, but 13% (7/54) of LIL items self-leaked, 98% of DSCR items restated an upstream phase’s exact percent volume change, 85% of PJRF items either stated the patient’s actual survival duration directly (31%) or named the upstream RANO label (72%), and 98% of TCM items restated LIL’s volume-change percentage or RANO label. The leak-controlled item set (v3), built with the same auditor folded into the drafting retry loop (Section 5), cleared every phase with zero violations, which shows compliance with that auditor (whose rules the drafting loop was optimized against), not that no leakage remains.

Post-operative rebaselining. Building the leak-controlled item set surfaced a second issue upstream of leakage: the LIL phase’s volume-change baseline was, for 42 of 54 patients, a pre-operative scan, so the measured volume change mostly captured the surgical resection rather than the treatment response that the RANO rating (DSCR) is actually judged against. We recompute the LIL baseline as each patient’s latest imaged post-operative scan prior to the follow-up (advancing past any re-resection that occurs before the follow-up date), changing the baseline for 44 of 54 patients and shifting the cohort’s volume-change direction from

a resection-dominated skew to a more balanced 20 patients increasing versus 32 decreasing. Two patients (Patient-049, Patient-068) lack a post-operative scan with matching segmentation and are excluded from the leak-controlled item set rather than mixed in on a pre-operative baseline, leaving 52 patients. Downstream DSCR, PJRF, and TCM facts and answer keys, which do not depend on the LIL baseline, are unchanged by this step; this also shows that the answer keys of later phases are not functions of earlier phases’ keys.

Difficulty of v3 relative to v2. v3 items are substantially harder than v2 for every model (image-present overall accuracy is 9–18 percentage points lower, e.g., Llama-4-Scout 66.7%→48.8%, MedGemma-4B 45.2%→34.2%; the two item sets cover 54 and 52 patients, so this is not a strictly paired comparison), consistent with the stem-leakage fix removing an exploitable shortcut rather than removing task difficulty. This drop is a validity correction to the earlier item set, not itself evidence about image use; we report it only to establish that the narrow with/without-image gap on v3 is not an artifact of v3 being an easier, less discriminating test.

C Metric Definitions

C.1 Vocabulary and Consistency Metrics

Answer recoverability. After a model answers, we issue a second call, the recoverability probe, containing *only* the model’s own reasoning text (no image, no question, no answer options) and ask which answer that reasoning supports. If the probe’s predicted answer matches the model’s original answer, the response is *recoverable*: the stated reasoning is sufficient, on its own, for a down-

stream reader (here, the probe model) to reconstruct the same conclusion. A mismatch means the model’s justification is disconnected from the answer it actually gave. This establishes textual self-consistency between an answer and its stated reasoning; it does not establish that the reasoning caused the answer, since a response can restate its own conclusion in different words without that restatement having driven the decision. We separately flag *correct-but-unrecoverable* responses, a right answer paired with reasoning that does not support it, as the most clinically dangerous category, since such a response offers no evidence it will generalize to a case that differs even slightly from the one it was scored on.

Ontology term coverage. This is KAB’s central grounding-adjacent metric; despite the name, it measures term *validity*, not correctness (see Section 8). Every diagnostic phase has an authoritative source of real clinical vocabulary: RadLex (Lan-glotz, 2006) for radiology anatomy, imaging modalities, and lesion descriptors (routed to the AIA and LIL phases), and the NCI Thesaurus (NCIt) for tumor classification, WHO grading, molecular markers, staging, and treatment (routed to DSCR, PJRF, and TCM). NCIt was chosen over UMLS because OmniBrainBench and LUMIERE are both predominantly glioma/glioblastoma datasets, for which NCIt has the deepest freely-available coverage, and because it requires no account or API key, removing a reproducibility barrier. We concatenate a model’s `visual_grounding` and `reasoning` fields, tokenize the text (preserving hyphenated compounds such as *ring-enhancing* as single tokens), and run a greedy longest-match scan against the union of both knowledge bases, trying 4-grams down to 1-grams with no reuse of consumed token positions. This yields `extracted_concepts`: every KB term the model used regardless of phase. We then take `matched_concepts` as the subset of `extracted_concepts` present in the phase-appropriate KB, and compute

$$\text{OntCoverage} = \frac{|\text{matched_concepts}|}{|\text{extracted_concepts}|}. \quad (1)$$

A low score indicates the model is not using vocabulary appropriate to the current diagnostic phase, for example still describing anatomy at the diagnosis phase rather than committing to tumor-classification language; unrecognized assertions are excluded from the denominator entirely, so this

score says nothing about whether the model’s *unrecognized* content is accurate, only about the recognized fraction.

Answer-term overlap. Ontology term coverage measures whether a model used *real* clinical terms; answer-term overlap measures whether it used ones that also appear in *this case’s correct answer text* — a proxy for on-target vocabulary, not for whether the surrounding explanation is a genuinely correct account of the case. Of the matched phase-KB concepts, we compute the fraction that also appear in the correct answer text:

$$\text{AnswerTermOverlap} = \frac{|\text{on-target matches}|}{|\text{matched_concepts}|}. \quad (2)$$

A model can score high on ontology term coverage while scoring low on answer-term overlap: using plausible-sounding real clinical language that is nonetheless wrong for the case at hand. The converse is also possible: repeating answer-option wording verbatim scores high on this metric without necessarily reflecting reasoning.

Upstream-term repetition. For each phase after AIA, we check what fraction of the key terms from the correct answers of *upstream* phases are referenced in the current phase’s reasoning text. This measures textual repetition of upstream terms, not demonstrated reasoning dependency: a model that identified a lesion in the right temporal lobe during LIL but never references that location during DSCR is *at least* answering each phase in isolation rather than visibly carrying context forward, but the converse does not hold, repeating an upstream term does not by itself show the current answer was reasoned *from* it, nor does omitting one show independent reasoning.

C.2 Feedback-Based Adaptation

The metrics above establish whether a model’s stated reasoning uses valid, on-target clinical vocabulary and is self-consistent with its answer, not whether the model reasoned correctly; the adaptation loop asks a different question: whether a model with low ontology term coverage can be corrected when the missing vocabulary is handed to it directly. The loop triggers when a response has low ontology term coverage (OntCoverage below a threshold τ_g , default 0.3) *and* has a non-empty set of missed concepts, defined as phase-KB terms present in the correct answer but absent from the model’s output. When triggered,

we re-issue the original prompt augmented with the model’s prior answer and reasoning and an explicit CLINICAL KNOWLEDGE CORRECTION block listing the missed concepts, and ask the model to re-examine the image in light of this correction. We record whether the model changed its answer (`adaptation_changed`) and whether the revised answer is correct (`adaptation_correct`), and report the *adaptation rate*: the fraction of triggered questions that the model corrects to the right answer. We describe this as performance under a targeted, answer-derived hint; correcting after such a hint does not by itself establish that the model already possessed the relevant knowledge and merely failed to activate it, since the hint also supplies new information. Distinguishing a genuine *knowledge gap* from an *activation failure* would require comparing this targeted-hint condition against a generic-retry control (re-asking without new information) and an irrelevant-hint control (a correction block naming concepts not relevant to the case); we have not yet run these controls, so we report the targeted-hint adaptation rate alone rather than the knowledge-gap/activation-failure distinction.

D OmniBrainBench Compatibility Details

The KAB framework as described in Section 3 presupposes that a case’s five phases represent one patient’s actual clinical trajectory: a model that fails AIA for a given patient should be gated out of that same patient’s DSCR question, not an unrelated patient’s. We investigated whether OmniBrainBench (Peng et al., 2026) supports this premise, by checking whether any image is shared across questions tagged with different clinical phases within a grouped case. Shared images are the evidence of continuity that the released data let us verify directly; they are not the only conceivable evidence, and the absence of shared images is not proof that no two questions concern the same patient.

OmniBrainBench ships 9,527 total QA pairs; we use its 6,823-question *closed-ended* (multiple-choice) subset, since KAB’s gating and scoring require answer options, and exclude the remaining 2,704 open-ended pairs. Of these 6,823 questions, drawn from 15 source sub-corpora (including PubMedVision, BraTS2021, RSNA, ISLES2022, and VQA_RAD, among others), we found only 2 verifiable cross-phase patient links, i.e. patients whose images appear in questions from two differ-

ent phases, both from VQA_RAD, each contributing one AIA question and one LIL question (4 questions total across the 2 pairs). We repeated the analysis with a second, independent method (extracting real per-patient case identifiers by stripping known modality/view/slice suffixes from filenames on a per-corpus basis) and obtained the same result: 2 verifiable multi-phase same-patient links in the entire dataset (4 questions), none spanning three or more phases, none spanning all five. Because the release provides no complete patient identifier, we cannot rule out further true links that neither method detects; the counts are lower bounds on verifiable links, not a census of patients. The root cause is structural: each of the 15 source sub-corpora was constructed for one specific clinical task (e.g., BraTS for tumor characterization, RSNA for hemorrhage classification, ISLES for stroke), so a given real patient within a sub-corpus is essentially only ever asked one phase-type of question. `source_file` is a field present in OmniBrainBench’s own released data, denoting each question’s origin annotation batch; using it to represent same-patient identity for a multi-phase causal chain is our own KAB-side interpretation, not a claim OmniBrainBench itself makes about that field. Under that interpretation, `source_file` in practice groups together an entire origin sub-corpus, standing in for a single patient; what this grouping evaluates as a chain is stitched from *different real patients’* questions, not one patient’s actual diagnostic journey. A model gated out of a later phase after failing an earlier one under this grouping demonstrates nothing about reasoning across any single patient’s chain. We frame this as an incompatibility between the released benchmark and our proposed sequential evaluation, not as a defect in a promise OmniBrainBench made: we are not aware of an explicit claim by its authors that its questions form patient-linked longitudinal chains.

We searched for a replacement or supplementary dataset with genuine same-patient, multi-phase continuity and a ready-made question layer (Section 2); none exists. This motivates the construction described next.

E Evidence-Audit Details and Option-Only Baselines

Evidence audit. Necessity and use are different claims: an image could be unnecessary on average while still being consulted, so we separately audit

the content of a model’s stated evidence against the same structured per-patient facts used to draft the items (Section 5). For each checkable claim in a response’s `visual_grounding` and reasoning fields (hemisphere/region statements and any measurement with units), we record two separate properties. *Availability*: whether the claim’s content already appears in the stem, answer options, or upstream chain context the model was shown. *Record status*: whether the claim is *supported* by the ground-truth facts (a measurement within 2% of a fact value, or a hemisphere present in the segmentation), *contradicted* (a hemisphere opposite to the segmentation, under the assumption that image left corresponds to patient right), *unmatched* (a measurement matching no fact value; pattern matching cannot show that such a value is wrong, since it may be a derived quantity), or *unverifiable* (a hemisphere claim with no segmentation to compare). Keeping the dimensions separate matters: an earlier single-label scheme reported any claim already present in shown text as “copied”, which hid the case in which a wrong claim is repeated from an earlier turn, and it read “not copied” as independent support. Responses with no checkable claim are reported separately, and claim-level denominators are stated. We treat hemisphere mismatches with care because the rendered slices follow radiological display convention without stating so in the prompt for the initial item set (fixed for the leak-controlled set, Section 5): a hemisphere claim that is wrong under one convention and right under the other reflects a display ambiguity, not a perceptual error. RANO-criteria thresholds quoted verbatim from clinical guidelines (e.g., “ $\geq 25\%$ increase”) are excluded, since they restate a rule rather than assert a patient-specific fact. The audit is deterministic pattern matching, not an LLM judge: numeric claims are compared by value against the facts and the shown-text pool, and hemisphere claims against the segmentation’s own left/right extent. This makes it reproducible from a response’s text, but it does not interpret paraphrase or synonymy (Section 8).

Option-only baselines. Guessers that use only the options and stem, computed on the v3 items exactly as shown (options shuffled; $n = 52$ per phase), are informative about answer-key artifacts. Always choosing the longest option scores 37% on AIA, 25% on LIL, 67% on DSCR (matching the majority class, since progressive disease is typically the longest option), 19% on PJRF, and 6% on TCM.

The most common correct letter, fitted on the same items and therefore optimistic, scores 31–37% on every phase. Choosing the option with the most words shared with the stem scores 0–33%. These sit at or below the four-option chance level except DSCR’s length effect, which the models do not exceed. AIA’s correct answer is the same string in every item, so a baseline that recognizes that text rather than a letter would score 100%. Two LLM baselines, an options-only condition and a no-chain-context condition, need model runs and are not yet reported.